

1. Mg–Ca–K triangle

WHAT YOU SEE

Top-left triangle linking Mg, Ca, and K.

WHAT IT ENCODES

Magnesium sits at the apex governing both calcium and potassium homeostasis, so a low Mg drags both down. Mechanistically, Mg is a required cofactor for PTH secretion AND for renal potassium retention; when Mg falls, PTH release stalls (→ hypocalcemia) and ROMK channels open unopposed (→ renal K wasting → hypokalemia). Normal serum Mg is roughly 1.7–2.4 mg/dL; symptoms cluster below ~1.2 mg/dL.

BOARD TRAP

When hypocalcemia OR hypokalemia 'won't correct' despite replacement, the trap is to keep up-dosing Ca or K — the unfixed low magnesium is blocking both, so you must replace Mg first.

2. Refractory hypo-K / hypo-Ca → check Mg (board)

WHAT YOU SEE

Chalkboard: 'REFRACTORY HYPOKALEMIA — CHECK Mg; REFRACTORY HYPOCALCEMIA — CHECK Mg'.

WHAT IT ENCODES

Hypomagnesemia must be corrected before potassium or calcium can be normalized — this is the single highest-yield teaching point of the scene. Low Mg perpetuates hypokalemia (unrestrained ROMK-mediated K secretion) and hypocalcemia (impaired PTH secretion + skeletal/renal PTH resistance), so both electrolytes 'leak' until Mg is restored. Give 1–2 g IV magnesium sulfate for symptomatic/severe cases.

BOARD TRAP

The classic exam vignette: a patient on a PPI or loop diuretic with persistent hypokalemia gets bag after bag of KCl with no improvement. Repleting K (or Ca) alone fails when Mg is low — checking and correcting magnesium is the answer.

3. Mg runs the tavern (banner)

WHAT YOU SEE

Top-right banner with Mg linked to Ca and K: 'Mg RUNS THE TAVERN'.

WHAT IT ENCODES

Magnesium is the master regulator: it is essential for PTH secretion (calcium control) and for renal tubular potassium handling, so any Mg disturbance ripples outward to Ca and K. Both extremes matter — low Mg suppresses PTH and wastes K, while very high Mg activates the CaSR and suppresses PTH too. About 99% of body Mg is intracellular/skeletal, so a 'normal' serum level can still hide total-body depletion.

BOARD TRAP

Trusting a normal-ish serum magnesium to exclude deficiency is the trap — because Mg is mostly intracellular, serum levels lag true depletion; in refractory hypocalcemia/hypokalemia, replete empirically rather than relying on a borderline serum value.

4. Sleeping pig — low Mg impairs PTH (sleeping)

WHAT YOU SEE

A sleeping pig 'ZZZ — NO Mg / Mg LOW'.

WHAT IT ENCODES

LOW magnesium SUPPRESSES PTH secretion (Mg is the cofactor the parathyroid needs to release PTH), producing a functional hypoparathyroidism with hypocalcemia. Paradoxically, while chronic mild Mg depletion stimulates PTH, SEVERE acute hypomagnesemia paralyzes PTH release entirely. Calcium will not recover until Mg is repleted — calcium and calcitriol alone fail.

BOARD TRAP

The resulting low Ca with low/normal PTH looks exactly like primary hypoparathyroidism — the trap is starting lifelong calcitriol/calcium instead of recognizing the reversible low-Mg cause. Always measure magnesium in any unexplained hypocalcemia.

5. PTH stuck / PTH resistant receptor

WHAT YOU SEE

A receptor: 'PTH STUCK / TARGET RECEPTOR / PTH RESISTANT'.

WHAT IT ENCODES

Beyond blocking PTH secretion, low magnesium also induces end-organ PTH RESISTANCE — bone and kidney become less responsive to whatever PTH is present, compounding the hypocalcemia. So hypomagnesemia hits calcium from two directions: less PTH released AND less PTH heard. This double hit is why hypocalcemia from low Mg is so refractory to calcium alone.

BOARD TRAP

Even if a PTH level comes back 'normal' in a hypomagnesemic patient, do not exclude Mg as the cause — the receptor resistance means a normal PTH is still ineffective, and only Mg repletion restores calcium responsiveness.

6. Central PTH gland (Mg activates CaSR)

WHAT YOU SEE

A muscular 'Mg' bartender labeled 'CENTRAL / Mg ACTIVATES CaSR'.

WHAT IT ENCODES

At the OTHER extreme, severe HYPERmagnesemia binds and activates the calcium-sensing receptor on the parathyroid (Mg mimics Ca), shutting off PTH and lowering serum calcium. This is why both ends of the magnesium spectrum disturb calcium — low Mg by removing the PTH-secretion cofactor, high Mg by tricking the CaSR into thinking calcium is high.

BOARD TRAP

A U-shaped relationship is the trap students miss: assuming 'more magnesium is always safe.' Hypermagnesemia (from infusions or renal failure) can itself suppress PTH and cause hypocalcemia plus neuromuscular/cardiac toxicity.

7. Kidney K-wasting (left character)

WHAT YOU SEE

A sad kidney: 'KIDNEY — K WASTING / REFRACTORY HYPOKALEMIA'.

WHAT IT ENCODES

Hypomagnesemia causes renal potassium wasting: normally intracellular Mg blocks the ROMK channel in the distal nephron, so when Mg is low ROMK is disinhibited and secretes potassium into the urine unchecked. The result is hypokalemia that resists KCl replacement. This is the mechanistic basis of the board mantra 'refractory hypokalemia → check Mg.'

BOARD TRAP

Continuing to push potassium replacement in a hypomagnesemic patient is futile and the trap — the unblocked ROMK channel keeps wasting the K you give until magnesium is restored to re-block it.

8. Mg high — suppressed / overpowered Ca

WHAT YOU SEE

Right side: 'Mg HIGH — Ca SUPPRESSED / OVERPOWERED, BP DOWN'.

WHAT IT ENCODES

Hypermagnesemia suppresses PTH (via CaSR activation) and acts as a physiologic calcium-channel/neuromuscular blocker, causing hypotension, flushing, nausea, and bradycardia at moderate levels (~4–6 mg/dL). It blunts the cardiovascular system, which is exactly why IV magnesium is used therapeutically to lower blood pressure and prevent seizures in pre-eclampsia.

BOARD TRAP

Mistaking early hypermagnesemia (hypotension, sluggish reflexes) for sepsis or dehydration is the trap — in a patient on a Mg infusion or with renal failure, check the Mg level; the treatment is stopping Mg and giving IV calcium gluconate as the antidote.

9. Hypermagnesemia: areflexia / respiratory depression

WHAT YOU SEE

A slumped patient: 'AREFLEXIA / RESPIRATORY DEPRESSION'.

WHAT IT ENCODES

Magnesium toxicity progresses by level: loss of deep tendon reflexes (~4–6 mg/dL) is the EARLY warning, then respiratory depression (~10 mg/dL), then cardiac arrest/asystole (>12–15 mg/dL). DTR monitoring is the bedside tool during therapeutic Mg infusions (eclampsia). IV calcium gluconate is the immediate antidote that antagonizes Mg at the neuromuscular junction.

BOARD TRAP

In eclampsia management, the trap is monitoring serum Mg levels alone — loss of deep tendon reflexes is the earliest clinical sign of toxicity and should prompt holding the infusion BEFORE respiratory depression develops.

10. Hypermagnesemia causes: IV infusion / renal failure / antacid-laxative

WHAT YOU SEE

Box: 'IV INFUSION, RENAL FAILURE, ANTACID/LAXATIVE, OVER-SUPPLEMENTATION'.

WHAT IT ENCODES

Hypermagnesemia almost always requires either a big Mg load or impaired excretion: IV magnesium infusion (eclampsia), advanced renal failure (can't excrete Mg), and Mg-containing antacids/laxatives (milk of magnesia, magnesium citrate) — the last especially dangerous when given to CKD patients. Healthy kidneys excrete excess Mg readily, so isolated hypermagnesemia in a normal-kidney patient is rare.

BOARD TRAP

Giving a magnesium-containing laxative or antacid to a CKD/dialysis patient is the trap — their kidneys cannot clear the load and they can develop life-threatening hypermagnesemia from an over-the-counter product.

11. GI losses — diarrhea / malabsorption / NG suction

WHAT YOU SEE

A patient: 'DIARRHEA / MALABSORPTION / NG SUCTION'.

WHAT IT ENCODES

GI loss is a leading cause of hypomagnesemia: chronic diarrhea, malabsorption (celiac, short bowel, IBD), NG suction, and notably long-term proton-pump inhibitor use (PPIs impair intestinal Mg absorption — an FDA-warned cause). Lower-GI fluid is Mg-rich, so diarrhea depletes Mg more than vomiting does.

BOARD TRAP

Forgetting PPIs as a cause of refractory hypomagnesemia is a common trap — a patient on chronic omeprazole with unexplained low Mg, Ca, and K may simply need the PPI stopped (or switched) in addition to repletion.

12. Renal wasting (diuretics) cause

WHAT YOU SEE

A patient with pills: 'RENAL WASTING'.

WHAT IT ENCODES

Renal magnesium wasting is caused by loop and thiazide diuretics, aminoglycosides, amphotericin B, cisplatin, calcineurin inhibitors, and the inherited tubulopathies Gitelman and Bartter syndromes. These impair Mg reabsorption in the loop of Henle/distal tubule. Gitelman classically gives the trio of hypomagnesemia, hypokalemia, and hypocalciuria.

BOARD TRAP

Attributing a diuretic patient's hypokalemia solely to kaliuresis is the trap — loop/thiazide diuretics waste magnesium too, and the coexisting hypomagnesemia is often what makes the hypokalemia refractory to KCl alone.

13. Alcohol — refractory hypomagnesemia

WHAT YOU SEE

A patient with a bottle: 'REFRACTORY / ALCOHOL'.

WHAT IT ENCODES

Chronic alcoholism is a classic cause of hypomagnesemia via poor intake, diarrhea, vomiting, and direct alcohol-induced renal Mg wasting. The low Mg then drives refractory hypocalcemia and hypokalemia. Alcoholics are also at high risk for refeeding-type shifts and thiamine deficiency, so Mg repletion is part of the broader resuscitation.

BOARD TRAP

In an alcoholic with seizures or arrhythmia, the trap is treating the calcium or potassium alone — the underlying magnesium depletion must be corrected first, and unrecognized hypomagnesemia is a frequent reason their electrolytes won't normalize.

14. Refeeding shifts Mg, K, PO4

WHAT YOU SEE

Bottom-center: 'REFEEDING / DKA → Mg, K, PO4 SHIFT'.

WHAT IT ENCODES

Refeeding syndrome (carbohydrate reintroduction after starvation) and insulin therapy in DKA drive an insulin-mediated intracellular SHIFT of magnesium, potassium, AND phosphate, dropping all three serum levels rapidly. Phosphate falls most dramatically. Anticipate and replete these electrolytes when refeeding malnourished patients or treating DKA.

BOARD TRAP

The trap in refeeding is watching potassium alone — the simultaneous fall in phosphate (can cause respiratory failure) and magnesium is just as dangerous, so monitor and replace all three preemptively, not reactively.

15. Replace Mg first / Ca alone fails

WHAT YOU SEE

Box: 'Ca ALONE FAILS — REPLACE Mg FIRST', with Mg IV bag.

WHAT IT ENCODES

The single highest-yield action: in refractory hypocalcemia or hypokalemia, REPLACE MAGNESIUM FIRST. Calcium and potassium cannot normalize while Mg is low because Mg is required for PTH secretion/action (Ca) and for blocking renal K wasting (K). Use IV magnesium sulfate (1–2 g) for symptomatic/severe deficiency; oral magnesium for milder chronic depletion.

BOARD TRAP

The board's favorite distractor is to keep selecting 'give more calcium gluconate' or 'give more KCl' for the refractory patient — the correct answer is to identify and replace the low magnesium that is sabotaging both.

16. Antidote / supportive care for hypermagnesemia

WHAT YOU SEE

Bottom-right: 'STOP Mg SOURCE / Ca GLUCONATE IV / SUPPORTIVE'.

WHAT IT ENCODES

Management of symptomatic hypermagnesemia: (1) STOP the magnesium source (infusion, laxatives, antacids); (2) give IV CALCIUM GLUCONATE as the immediate physiologic antagonist that reverses neuromuscular/cardiac depression; (3) supportive care — IV fluids and loop diuretics to enhance renal excretion, and dialysis if severe or in renal failure.

BOARD TRAP

Forgetting that IV calcium is the antidote to magnesium toxicity is the trap — just as magnesium is calcium's antagonist, calcium gluconate immediately counteracts dangerous hypermagnesemia while you work to remove the magnesium.